

CAMBRAI EPINOY, France

WMO: 070170

Lat: 50.222N

Lon: 3.154E

Elev: 257

StdP: 14.56

Time Zone: 1.00 (EUC)

Period: 99-12

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.9	24.6	16.0	12.5	23.6	20.3	15.4	26.0	33.6	49.1	30.2	48.0	7.6	50	0.606

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.3	84.9	68.2	80.9	66.6	77.5	65.1	70.0	81.5	68.0	78.2	66.4	75.2	9.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.7	95.9	74.2	64.3	91.2	72.0	62.9	86.6	70.6	34.0	82.1	32.4	78.3	31.1	75.5	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.1	23.1	20.6	16.5	91.5	4.8	4.0	13.0	94.4	10.1	96.7	7.4	98.9	3.9	101.8		
(5)			15.9	73.2	4.8	1.9	12.5	74.6	9.6	75.7	6.9	76.7	3.5	78.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.5	38.2	39.8	44.8	50.3	56.3	61.3	64.5	64.5	60.5	53.1	45.1	38.5	(6)
(7)		DBStd	11.38	7.53	6.89	6.27	6.13	5.64	5.40	5.26	4.77	5.14	5.99	6.01	7.05	(7)
(8)		HDD50	1484	369	290	182	69	11	0	0	0	0	37	166	360	(8)
(9)		HDD65	5101	832	707	627	441	276	138	75	64	155	368	596	822	(9)
(10)		CDD50	2021	3	3	20	78	205	338	450	451	315	135	20	3	(10)
(11)		CDD65	162	0	0	0	0	5	26	60	50	20	1	0	0	(11)
(12)		CDH74	1566	0	0	0	16	84	285	550	458	164	9	0	0	(12)
(13)		CDH80	400	0	0	0	0	8	69	153	144	25	1	0	0	(13)
(14)	Wind	WSAvg	10.4	12.2	11.6	11.0	10.1	9.6	8.8	9.2	9.0	9.4	10.8	11.3	11.9	(14)
(15)	Precipitation	PrecAvg	28.7	2.3	2.1	2.0	1.9	2.3	2.6	2.6	2.8	2.2	2.6	2.6	2.9	(15)
(16)		PrecMax	39.2	5.7	4.2	6.7	4.9	4.9	5.0	4.4	6.6	4.9	5.6	4.7	7.2	(16)
(17)		PrecMin	22.0	0.2	0.5	0.2	0.1	0.2	1.1	0.9	0.4	0.5	0.5	0.7	0.9	(17)
(18)		PrecStd	4.4	1.1	1.1	1.3	1.1	1.2	1.1	1.0	1.5	1.2	1.1	1.1	1.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.2	56.7	66.1	76.3	80.2	87.3	89.1	91.6	83.4	72.0	61.0	55.4	(19)
(20)			MCWB	52.5	51.0	53.6	60.1	65.9	69.9	68.9	70.9	66.6	63.4	57.2	53.3	(20)
(21)		2%	DB	53.1	53.5	61.6	71.3	76.3	81.1	84.7	84.7	78.8	66.9	58.0	53.2	(21)
(22)			MCWB	51.0	49.8	53.0	57.9	62.6	66.4	68.2	68.3	65.7	61.1	55.2	51.5	(22)
(23)		5%	DB	50.9	51.5	57.7	65.5	72.4	76.9	80.4	79.5	74.3	63.8	55.8	51.1	(23)
(24)			MCWB	48.9	48.7	51.4	55.1	61.1	64.8	66.4	66.2	64.0	59.3	53.4	49.4	(24)
(25)		10%	DB	48.7	49.4	54.7	61.4	68.0	72.7	76.3	74.7	69.9	61.5	53.8	48.9	(25)
(26)			MCWB	46.9	47.0	50.1	53.3	58.9	62.6	64.9	63.8	61.9	57.8	51.7	47.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.5	52.5	56.1	62.1	67.8	72.5	71.4	71.8	69.5	64.6	57.6	53.8	(27)
(28)			MCD B	54.6	54.8	61.3	73.9	77.7	83.8	83.0	86.1	78.6	70.0	59.7	54.9	(28)
(29)		2%	WB	51.2	50.7	54.3	58.6	64.2	68.2	69.7	69.6	67.1	61.9	55.5	51.7	(29)
(30)			MCD B	52.7	52.7	59.3	67.2	72.3	78.2	81.6	81.8	76.0	65.8	57.4	53.0	(30)
(31)		5%	WB	49.1	49.0	52.6	56.3	61.9	65.9	67.7	67.4	65.0	60.0	53.8	49.5	(31)
(32)			MCD B	50.7	51.0	57.2	63.9	70.4	74.8	77.5	77.2	72.1	63.3	55.6	50.8	(32)
(33)		10%	WB	46.9	47.2	50.4	54.1	59.9	63.8	65.9	65.5	63.1	58.4	51.8	47.4	(33)
(34)			MCD B	48.4	49.1	54.0	60.2	67.1	71.2	74.7	72.7	68.5	60.9	53.6	48.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.5	10.2	13.9	17.0	17.2	17.7	17.3	17.1	16.6	13.1	9.9	8.1	(35)
(36)			MCDBR	9.2	12.4	20.0	24.1	24.6	24.5	24.5	25.4	22.8	14.9	11.2	8.9	(36)
(37)			MCWBR	8.5	9.5	12.5	13.2	13.1	12.3	10.6	11.0	11.5	9.6	9.0	8.7	(37)
(38)		5% WB	MCDBR	9.0	10.2	16.8	20.8	21.7	21.2	21.3	21.7	19.5	13.2	10.5	8.7	(38)
(39)			MCWBR	8.5	8.8	11.0	12.2	12.4	11.9	10.2	10.3	10.5	9.1	8.9	8.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.327	0.348	0.384	0.410	0.420	0.418	0.418	0.408	0.382	0.364	0.338	0.322	(40)	
(41)		taud	2.393	2.357	2.247	2.191	2.207	2.238	2.238	2.289	2.348	2.397	2.398	2.367	(41)	
(42)		Ebn at Noon	223	248	257	263	266	267	265	261	256	238	216	205	(42)	
(43)		Edh at Noon	21	27	36	42	43	42	42	38	32	26	20	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	266	490	862	1360	1586	1746	1656	1412	1083	636	313	213	(44)	
(45)		RadStd	28	90	113	152	148	162	163	136	84	54	33	30	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (63 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page